# Exam #3 (Sta-209, Sp24, 5/10/24)

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**Directions**

- Answer each question using *no more than specified number of sentences* and not attempt to avoid these guidelines by using run-on sentences. Answers that are unnecessarily verbose may result in point loss.
- Do not include superfluous information in your answers, you may be penalized if you make an inaccurate statement even if you go on to provide a correct answer.

**Formula Sheet**

**Definitions**:

- **Risk**: relative frequency of an event/outcome
- **Relative Risk**: ratio of the risks across two groups
- **Odds**: ratio of how often an event/outcome is observed relative to how often it is not observed
- **Odds ratio**: ratio of odds across two groups

**Formulas**:

$$\bar{x} = \tfrac{1}{n}\sum_{i=1}^{n} x_i$$

$$s = \sqrt{\tfrac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

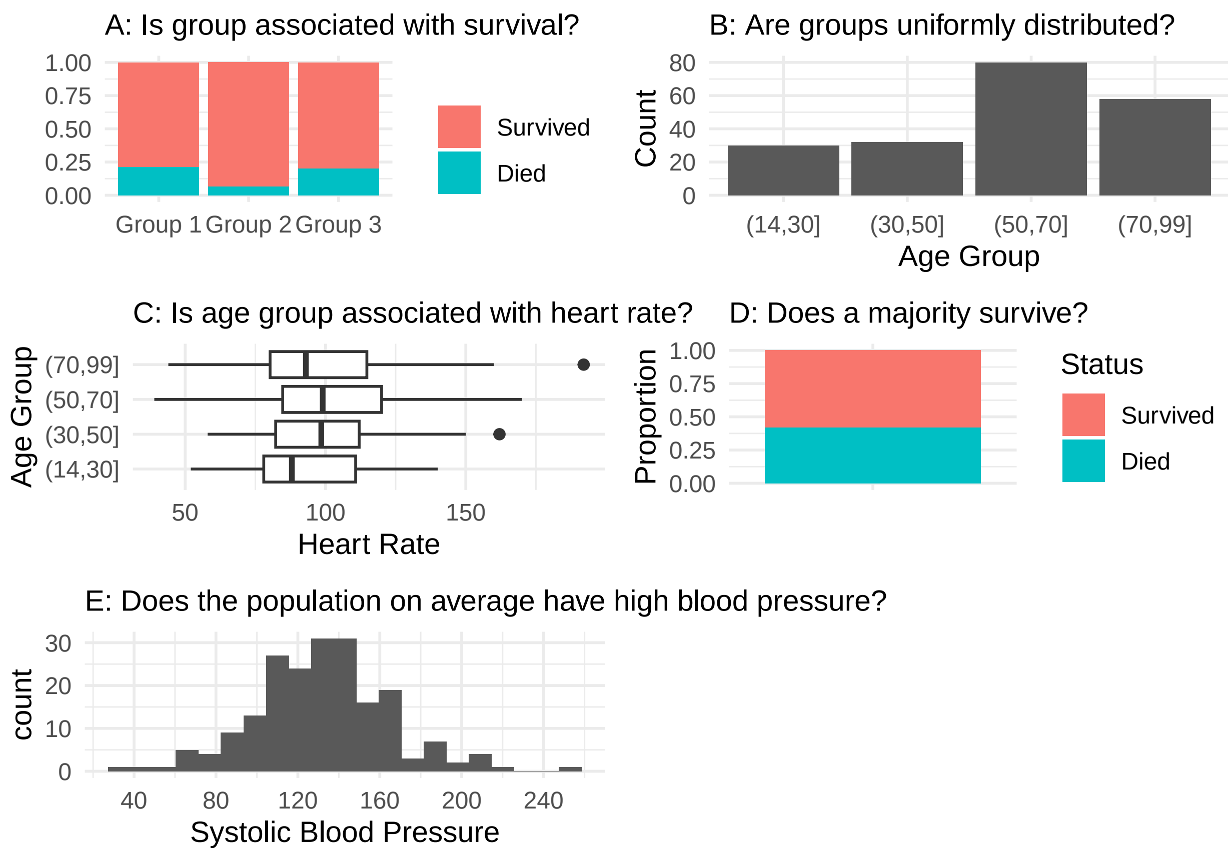
$$r = \tfrac{1}{n-1}\sum_{i=1}^{n}\left(\tfrac{x_i-\bar{x}}{s_x}\right)\left(\tfrac{y_i-\bar{y}}{s_y}\right)$$

| Statistic | Standard Error | Conditions |
|---|---|---|
| $\hat{p}$ | $\sqrt{\frac{p(1-p)}{n}}$ | $np \geq 10$ and $n(1-p) \geq 10$ |
| $\bar{x}$ | $\frac{\sigma}{\sqrt{n}}$ | normal population or $n \geq 30$ |
| $\hat{p}_1 - \hat{p}_2$ | $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$ | $n_ip_i \geq 10$ and $n_i(1-p_i) \geq 10$ for $i \in \{1,2\}$ |
| $\bar{x}_1 - \bar{x}_2$ | $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ | normal populations or $n_1 \geq 30$ and $n_2 \geq 30$ |

Chi-squared test statistic: $X^2 = \sum_{j=1}^{k} \frac{(\text{observed}_j - \text{expected}_j)^2}{\text{expected}_j}$

## Question #1

**Part A**: For each of the following data visualizations provide the name of a hypothesis test that could be used to assess whether the displayed results could be due to sampling variability. Be sure to read the title of each graph and use its label (ie: "A", "B", etc.) to reference it. You do not need to explain your answer, you only need to provide the name of the test.




Graph A:

Graph B:

Graph C:

Graph D:

Graph E:

**Part B**: Consider using an F-test to compare the fit of the following linear regression models, Model A: Income $= b_0 + b_1 * \text{Years Education} + b_2 * \text{IQ Score} + \epsilon$, and Model B: Income $= b_0 + b_1 * \text{Years Education} + \epsilon$. Are these models nested? If so, briefly explain what makes them nested; If not, provide a model that is nested with Model A.

**Part C**: Consider the use of one-way ANOVA to evaluate the relationship between the response variable *survival time* after being diagnosed with cancer and the explanatory variable race/ethnicity (using the US Census's 7 categories of White, Hispanic/Latino, Black, Asian, American Indian, Middle Eastern or North African, and Pacific Islander). Suppose all assumptions of the test are met and the $p$-value of the test is 0.02, which prompts your friend to conclude that White individuals have different average survival times than Black individuals. Do you agree with your friend's conclusion? State "agree" or "disagree" and briefly explain your reasoning.

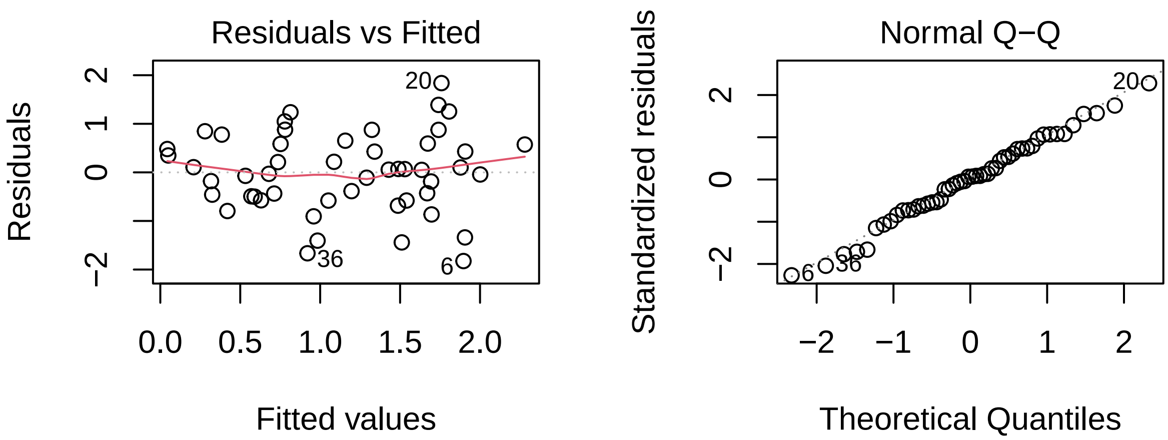
**Part D**: The following statements are either True or False. For each clearly state "True" or "False" and if a statement is False provide a brief explanation (1-sentence) describing what makes the statement incorrect.

- i: One-way ANOVA requires a Normally distributed response variable in order for the test's $p$-value to be reliable.

- ii: Chi-squared tests are used in scenarios involving nominal categorical variables.

- iii: An F-test can be used to test for an association between a categorical explanatory variable and a quantitative outcome while adjusting for a quantitative confounding variable.

- iv: The residual plots displayed below suggest that hypothesis testing results for the corresponding linear regression model are reliable:




- v: If we use an F-test to compare two regression models and fail to reject the null hypothesis the best practice is to favor the model that uses more explanatory variables.

## Question #2

We've previously analyzed the "flicker" data set, where researchers measured the critical flicker frequency (the variable `Flicker`), which is the highest frequency where a person can differentiate between a solid and flickering light source, for subjects with three different eye colors: blue, brown, and green (the variable `Color`). The researchers hypothesized that eye color was associated with critical flicker frequency.

Below are some descriptive statistics from this study that you should use throughout this question:

```
## Overall summary
flicker %>%
  summarize(mean_flicker = mean(Flicker),
            median_flicker = median(Flicker),
            sd_flicker = sd(Flicker),
            n = n())
```

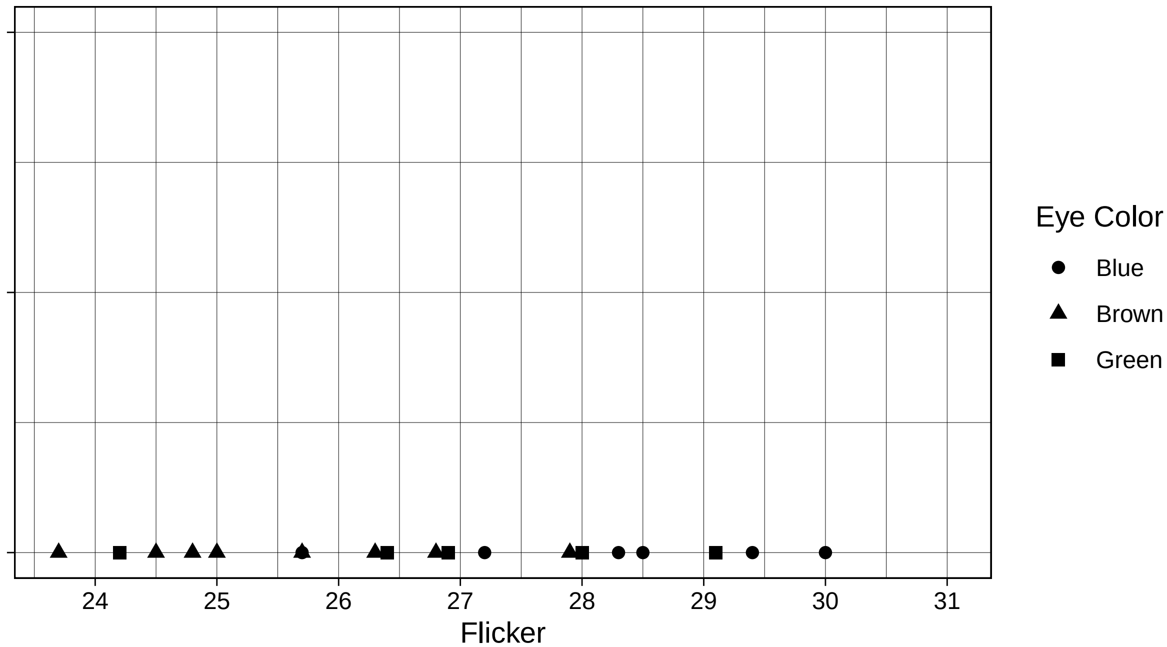
```
##   mean_flicker median_flicker sd_flicker  n
## 1     26.75263           26.8   1.845526 19
```

```
## Summary by eye color
flicker %>%
  group_by(Color) %>%
  summarize(mean_flicker = mean(Flicker),
            median_flicker = median(Flicker),
            sd_flicker = sd(Flicker),
            n = n())
```

```
## # A tibble: 3 x 5
##   Color mean_flicker median_flicker sd_flicker     n
##   <chr>        <dbl>          <dbl>      <dbl> <int>
## 1 Blue          28.2           28.4       1.53     6
## 2 Brown         25.6           25.4       1.37     8
## 3 Green         26.9           26.9       1.84     5
```

**Part A**: One-way ANOVA can be expressed as a comparison of statistical models. With this in mind, briefly describe the *null model* using either words or statistical symbols (or both).

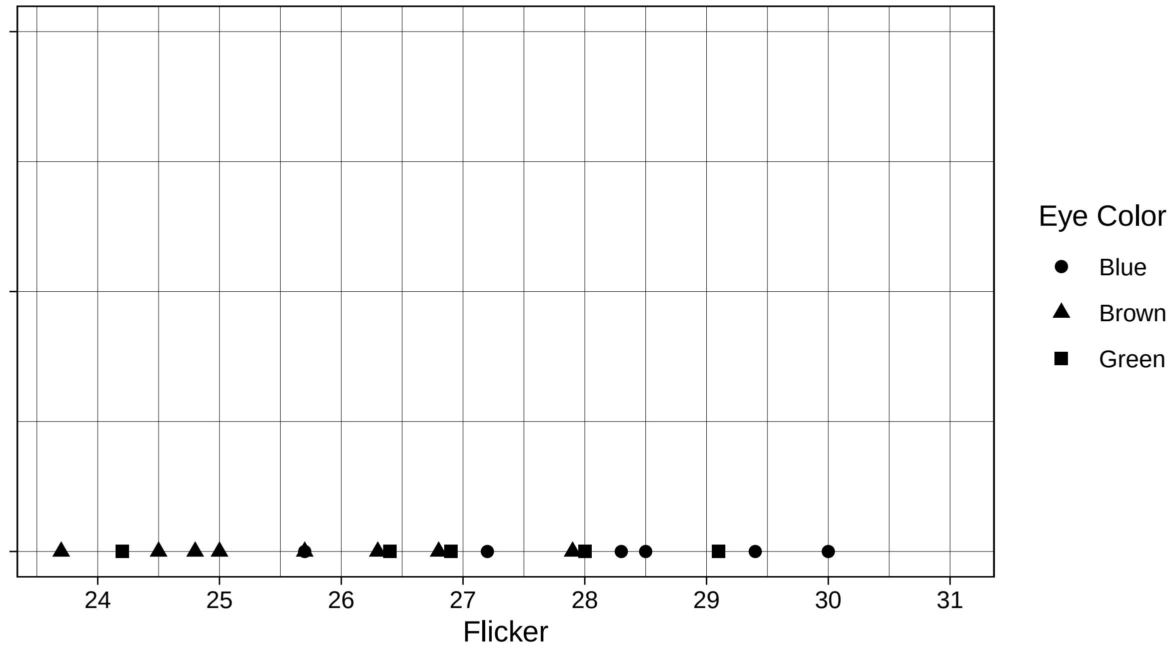
**Part B**: The *null model* in one-way ANOVA involves using a certain distribution to model the data. Sketch this distribution on the graph below. Be careful where you place the center of the distribution.




**Part C**: One-way ANOVA compares models using sums of squares, which involve the squared deviations between observed and predicted values (ie: squared residuals). More specifically, $SS = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. With this is in mind, what is the *deviation/residual* for the observation with the highest observed critical flicker frequency *under the null model*. Use the data shown on the x-axis of the graph given above.

**Part D**: Now briefly describe the *alternative model* involved in one-way ANOVA using either words or statistical symbols (or both).

**Part E**: Similar to Part B, sketch the *alternative* model involved in one-way ANOVA on the graph given below:




**Part F**: Similar to Part C, what is the *deviation/residual* for the observation with the highest observed critical flicker frequency *under the alternative model.*

**Part G**: The sum of squared residuals for the null model, $SST$, in this application is 61.31. Do you expect the sum of squared residuals for the alternative model to be larger, smaller, or approximately equal to this value? State "larger", "smaller", or "approximately equal" and briefly explain your reasoning.

**Part H**: The F-statistic and $p$-value for the one-way ANOVA that analyzes the relationship between `Color` and `Flicker` are $F = 4.8$ and $p = 0.023$ respectively. Based upon these results, provide a brief conclusion in regard to the researcher's hypothesis described in the introduction of Question 2. You may assume the conditions of the test have been met.

**Part I**: Shown below are the results of post-hoc pairwise testing for the ANOVA model described throughout this question. Briefly explain how these results allow you to make a more precise conclusion than the conclusion you reached in Part H.

```
model = aov(Flicker ~ Color, data = flicker)
TukeyHSD(model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Flicker ~ Color, data = flicker)
##
## $Color
##                  diff        lwr        upr     p adj
## Brown-Blue  -2.595833 -4.7621317 -0.4295349 0.0181490
## Green-Blue  -1.263333 -3.6922387  1.1655720 0.3934460
## Green-Brown  1.332500 -0.9542388  3.6192388 0.3155339
```

## Question #3

Gregor Mendel conducted breeding experiments using *Pisum sativum*, a type of pea plant, in order to understand heritable traits. Mendel's framework categorized the genetic profile of a heritable trait as either "AA", "Aa", or "aa".

Under this framework, if two parents who both have the "Aa" genotype are bred, then 25% of their offspring will have the "AA" genotype, 50% will have the "Aa" genotype, and 25% will have the "aa" genotype.

Suppose a biologist breeds two pea plants with unknown genotypes, but tests their offspring and finds the following distribution of genotypes:

- 10 "AA"
- 24 "Aa"
- 7 "aa"

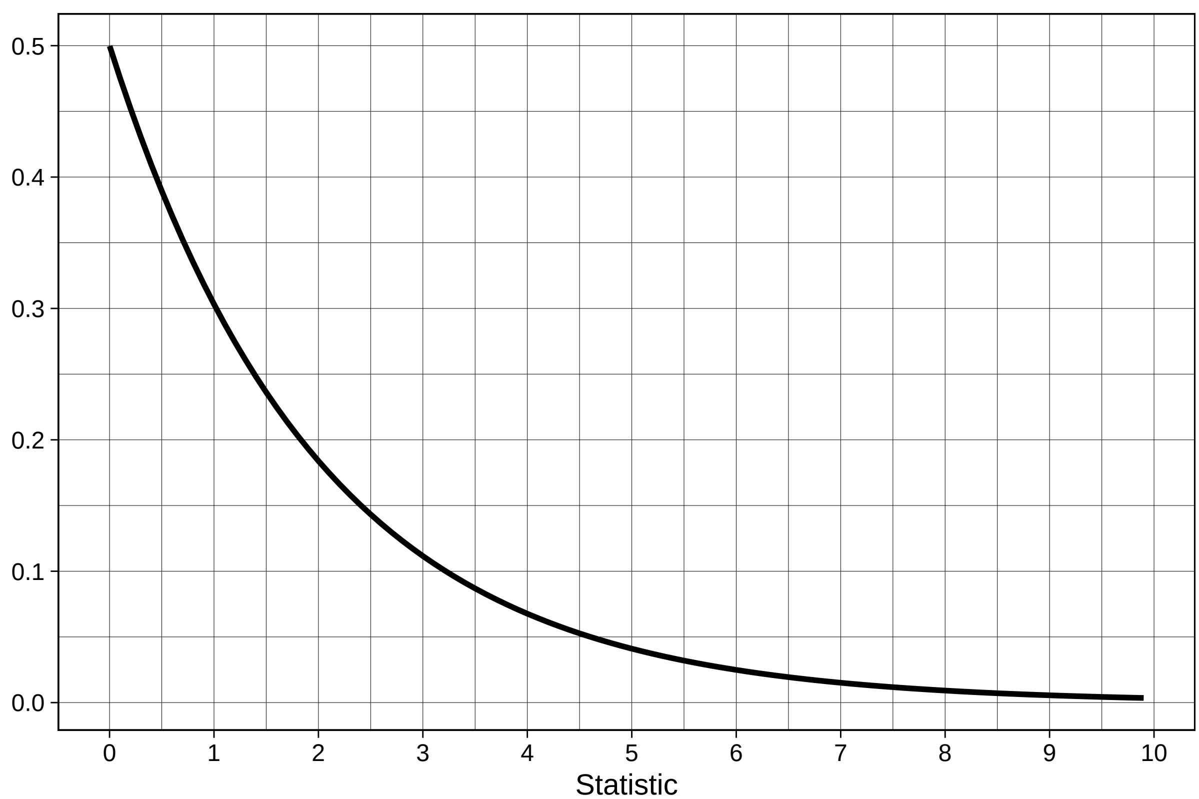
**Part A**: Suppose the biologist wants to use a statistical test to determine if one or more of the parent pea plants did not have the "Aa" genotype. Provide the name of the test they should use. You only need to provide the name of the test, nothing else.

**Part B**: Using either words or statistical symbols, provide the null hypothesis for the hypothesis test you named in Part A.

**Part C**: If the null hypothesis from Part B is true, how many plants of each genotype would the biologist expect to observe?

**Part D**: Calculate a test statistic that can be used to evaluate the null hypothesis from Part B. Show your calculation.

**Part E**: Shown below is the null distribution of the test statistic you calculated in Part D. Shade the portion of this graph that represents the *p*-value in this application and provide a brief conclusion based upon your estimate of the *p*-value from this graph. You do not need to perfectly estimate the *p*-value, but you should provide a conclusion that is consistent with your shading/estimation.




*Remember to provide your conclusion after shading the graph.*